

SpaceY

Safe Yields from Space Resources



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1 Executive Summary

Cosmic Quarry Mining Corporation is expanding its interstellar mining operations across the Helionis Cluster, Bayesia System, and Oryn Delta. These environments expose the company to four main operational hazards: Workers' Compensation, Cargo Loss, Equipment Failure, and Business Interruption. To address these risks, this report recommends that Galaxy General Insurance Company offer a full insurance portfolio covering all four hazard areas.

The recommendation is supported by both the risk assessment and the pricing analysis. Risk exposure differs materially across the three solar systems. Helionis has the highest Equipment Failure risk and elevated Cargo Loss exposure because of its large, aging machinery base and debris-related transit conditions. Bayesia is more exposed to environmental stress, leading to relatively higher Workers' Compensation and Business Interruption risk. Oryn Delta presents the most unstable operating environment overall, with the highest Cargo Loss and Business Interruption exposure. These differences support offering all four hazard lines with system-specific underwriting adjustments.

From a pricing perspective, the portfolio is commercially viable under both short-term and long-term views. Under the one-year baseline, aggregate expected loss is \$505.42 million, premium is \$631.77 million, and net revenue is \$126.35 million. Cargo Loss is the main driver of expected loss and portfolio risk, while Business Interruption contributes a heavier tail under stress. Over the 10-year horizon, the aggregate present-value expected loss is \$4,932.55 million, with a single premium of \$5,605.17 million and net revenue of \$672.62 million, based on exposure growth, inflation, discounting, and a lower long-term target return assumption. Stress testing shows that the portfolio remains supportable under adverse conditions, although stronger capital backing is needed in severe scenarios. Relative to baseline, required capital increases by 23.5% under the moderate scenario and by 50.9% under the worst scenario.

Overall, the recommendation is to implement the full four-product portfolio for an initial 10-year horizon. The proposed structure addresses CQM's main operational risks, supports broader enterprise risk management, and remains commercially sustainable for Galaxy General, while recognizing that some judgment is required because the new solar systems do not have directly comparable historical loss experience. Taken together, the results provide a sound basis for proceeding with a tailored and scalable insurance program for Cosmic Quarry's long-term expansion.

2 Product Design

Certain hazards require additional structural adjustments to reflect their risk characteristics. Workers' Compensation adopts a lower deductible to preserve meaningful worker protection and avoid excessive cost shifting to employees. Cargo Loss has a larger claim scale and heavier tail, so its coverage limit is set at VaR_{80} severity. Equipment Failure includes a specific sublimit for the *Quantum Bore*, reflecting concentrated tail risk from high-value machinery. Business Interruption arises from operational downtime rather than discrete damage events, and is therefore structured with a revenue-based waiting-period proxy and an annual aggregate cap to control accumulation risk. The four aggregate loss distribution plots shown in Figure 2.1 provide insights for the product design regarding these benefit structures, as they illustrate the portfolio loss profiles of the four hazard lines across the three solar systems.



Figure 2.1: Aggregate loss distributions by hazard before policy limit

The key policy design criteria are summarized in Table 2.1, while detailed numerical calibration of the benefit structures across hazards and solar systems is provided in Appendix A.

Table 2.1: Benefit structure by hazard

Hazard	Deductible	Limit	Other Feature
WC	VaR ₅₀ severity	VaR ₉₅ severity	–
CL	VaR ₆₀ severity	VaR ₈₀ severity	–
EF	VaR ₆₀ severity	VaR ₉₅ severity	Quantum Bore sublimit (VaR ₉₀ severity)
BI	Waiting-period proxy	VaR ₉₀ severity	Annual aggregate cap (VaR ₉₉)

Notes: WC = Workers’ Compensation; CL = Cargo Loss; EF = Equipment Failure; BI = Business Interruption. These abbreviations are used hereinafter throughout the report. Waiting-period proxy = BI deductible $\approx$ daily revenue $\times$ 2

Coverage Trigger and Exclusions

Coverage triggers: Coverage is triggered when an insured operational event causes loss above the applicable deductible or waiting-period threshold.

EF: sudden machinery breakdown from debris impact, asteroid collision, or radiation disturbance; CL: physical loss or damage to insured cargo during authorized transit due to debris collision, container failure, or navigation disruption; BI: measurable revenue loss following partial or complete operational suspension caused by an insured disruption, subject to the waiting period; WC: employee injury or illness directly arising from mining activities, including equipment accidents and hazardous exposure.

Exclusions: wear and tear, gradual deterioration, inadequate maintenance, operation outside approved environmental or navigation limits, failure to maintain required protection or monitoring systems, intentional misconduct, safety breaches, regulatory non-compliance, unauthorized expansion, and pure market-driven revenue decline.

3 Summary of Pricing and Capital Modeling

The proposed coverages are assessed using a pricing framework that combines expected loss, expense loading, risk margin, capital loading, and target profit margin. The baseline claims structure is estimated from historical claims data (see Appendix B) and then mapped to

the new solar systems using the available operational, exposure, and system-specific risk information (see Appendix C). The fitted historical models underlying the pricing analysis are summarized in Table B.1.

In practice, loadings are applied through General Galaxy Insurance’s internal pricing framework rather than calculated separately. Consistent with internal pricing benchmarks, the product is targeted to achieve an expected loss ratio of 70%, allowing for expenses, capital costs, and target underwriting return. Therefore, premiums in this report are determined by applying the target loss ratio benchmark to the modeled expected losses.

Table 3.1: Summary of final frequency and severity models

Hazard	Frequency Model	Severity Model
Workers’ Compensation	Poisson GLM	Lognormal body + Weibull-type GPD tail
Cargo Loss	Quasi-Poisson GLM	Lognormal body + Fréchet-type GPD tail
Business Interruption	Negative Binomial GLM	Lognormal
Equipment Failure	Poisson GLM	Quantum Bore: Lognormal body + empirical bootstrap tail; other equipment: Lognormal

3.1 Aggregate loss distributions by hazard coverage

Based on the fitted frequency–severity models and the transfer adjustments for the new solar systems, we simulate aggregate insured losses for each hazard coverage under both short-term and long-term pricing views. The short-term analysis is based on one-year Monte Carlo simulation and is used to assess baseline expected loss, volatility, and tail risk across coverages. The long-term analysis extends the loss projections over the full protection horizon by incorporating system-specific exposure growth, inflation, and yield-curve discounting, and is used to evaluate single-premium pricing, net revenue, and target return on a present-value basis.

Table 3.2 reports the short-term baseline results. Cargo Loss (CL) is the dominant source of expected loss and portfolio risk, while Workers’ Compensation (WC) remains relatively small in both scale and tail exposure. Equipment Failure (EF) and Business Interruption (BI) occupy an intermediate position, with BI contributing a materially heavier tail than EF despite its lower expected loss. At the portfolio level, these results provide a baseline view of one-year underwriting exposure before introducing stressed or correlated scenarios.

Table 3.2: Short-term baseline pricing summary by hazard coverage

Metric	WC	CL	EF	BI	Aggregate
Expected Loss	0.56	465.18	16.17	23.50	505.42
Standard Deviation	0.05	41.06	1.20	22.51	46.86
VaR ₉₉	0.75	641.72	20.97	150.71	814.16
Premium	0.71	581.48	20.22	29.38	631.77
Return	0.20	0.20	0.20	0.20	0.20
Net Revenue	0.14	116.30	4.04	5.88	126.35

Notes: Aggregate refers to the portfolio-level result across all four hazard coverages. All amounts are reported in millions of dollars.

Table 3.3 presents the long-term baseline view. These results reflect projected exposure growth across solar systems, inflation adjustment, and discounting using the risk-free spot-rate term structure. As in the short-term results, Cargo Loss remains the largest contributor to expected loss and overall portfolio risk. For long-term pricing, we adopt a lower target return of 12%, which is more appropriate for a multi-year relationship and reflects both the strategic value of a stable client portfolio and the protection already provided by deductibles, limits, and sublimits. The resulting single-premium estimates provide a present-value benchmark for long-horizon underwriting and profitability assessment.

Table 3.3: Long-term baseline pricing summary by hazard coverage

Metric	WC	CL	EF	BI	Aggregate
Expected Loss	5.57	4573.33	126.61	227.05	4932.55
VaR ₉₉	6.64	5550.70	151.53	960.76	6100.88
Single Premium	6.33	5196.96	143.87	258.01	5605.17
Net Revenue	0.76	623.64	17.26	30.96	672.62
Return	12.0%	12.0%	12.0%	12.0%	12.0%

Notes: Long-term results reflect exposure growth, inflation, and yield-curve discounting. Monetary quantities are reported in present-value millions. Single premium is set to achieve a target return rate of 12%, where return is defined as net revenue divided by single premium. Hazard-level VaR₉₉ values are approximate if obtained by nominal-to-present-value scaling, while the aggregate VaR₉₉ is calculated directly from the simulated portfolio distribution.

3.2 Stress testing for extreme scenarios

Using the three stress scenarios introduced in Section 4.3 and parameterized in Appendix D.1–D.2, we stress the baseline portfolio to assess pricing resilience and capital adequacy under adverse conditions. The scenarios are: (i) a *best / baseline* attritional case across all three systems, (ii) a *moderate* two-system disruption affecting Helionis Cluster and Oryn Delta, and (iii) a *worst* firm-wide correlated disruption affecting all three systems.

Table 3.4 compares the stressed portfolio with the baseline in terms of expected loss, tail risk, and required capital, where required capital is defined as VaR_{99} minus expected loss. Relative to baseline, the moderate scenario increases required capital by 23.5%, indicating reduced pricing margin and a need for additional capital support. Under the worst scenario, required capital rises by 50.9%, implying materially tighter underwriting, stronger pricing discipline, and greater capital backing.

Table 3.4: Stress testing results for pricing and capital

Scenario	EL	VaR_{99}	Required Capital	Δ Capital vs Base-line	Pricing / capital implication
Best / baseline	505.42	625.37	119.95	–	Baseline pricing benchmark
Moderate	605.27	753.42	148.15	23.5%	Margin compressed; additional capital buffer required
Worst	757.31	938.26	180.95	50.9%	Severe tail stress; capital support and tighter underwriting terms required

Notes: Results are reported at the aggregate portfolio level and in millions of dollars. Required Capital is defined as VaR_{99} minus Expected Loss.

4 Risk Assessment

4.1 Risk identification by hazard coverage and solar system

Risk exposure differs across the three solar systems because of differences in environment, operating scale, and infrastructure reliability.

- Helionis Cluster has the highest Equipment Failure risk because of its large, aging machinery base and high utilization. Cargo Loss is also elevated due to debris-related

transit exposure, while Workers' Compensation and Business Interruption are more moderate.

- Bayesia System is more exposed to environmental stress than to scale-driven risk. Radiation and electromagnetic activity increase Workers' Compensation and Business Interruption risk and place moderate pressure on equipment, while Cargo Loss is lower because transport routes are more standardized.
- Oryn Delta operates in the most unstable environment. Dense debris, rugged terrain, orbital shear, and weaker communications increase Cargo Loss, Workers' Compensation, and Business Interruption risk. Equipment Failure frequency is moderated by newer machinery, but severity remains elevated.

Table 4.1: Risk identification by hazard coverage and solar system

System	WC	CL	EF	BI
Helionis	3	4	5	3
Bayesia	4	2	3	4
Oryn	4	5	4	5

Notes: Risk scores are ordinal and range from 1 (very low) to 5 (very high).

These relative differences are summarized in Table 4.1, which assigns ordinal risk scores to each hazard line by solar system.

4.2 Correlated risk scenarios

Although Helionis Cluster, Bayesia System, and Oryn Delta operate in separate stellar systems, losses may still be correlated through common interstellar shocks and shared operational dependencies. Accordingly, correlated scenarios are defined through deep-space events and shared relay, transport, navigation, and regulatory channels, rather than through a single local stellar-weather event affecting all systems simultaneously. The main correlated scenarios considered in this report are summarized in Table 4.2.

Further hazard-level interpretation is provided in Appendix E.

Table 4.2: Summary of correlated risk scenarios across solar systems

Scenario	Systems	Channel	Main hazards	Score
Radiation shock	Bayesia + Oryn	Radiation / electronics	EF, BI, secondary WC	4
Relay failure	Helionis + Oryn	Communications	BI, CL, secondary EF	4
Navigation failure	Helionis + Oryn	Navigation / debris	CL, EF, BI	4
Transport shock	All three	Logistics / transport	CL, BI, EF, secondary WC	5

Notes: Scores are ordinal and range from 1 (very low) to 5 (very high), reflecting relative portfolio severity rather than probability.

4.3 Scenario testing

Based on the system-specific vulnerabilities and correlated risk channels identified earlier, three stress scenarios are considered: a *best* / *baseline* scenario, a *moderate* two-system disruption, and a *worst* firm-wide disruption. This framework distinguishes ordinary claims experience from multi-system stress and provides a basis for pricing, capital assessment, and underwriting judgment.

Table 4.3: Scenario testing summary

Scenario	Systems	Stressed hazards	Quantitative impact
Best	All three	Attritional only	EL = 505.42, VaR ₉₉ = 625.37
Moderate	Helionis + Oryn	CL, EF, BI	EL = 605.27, VaR ₉₉ = 753.42, Δ VaR ₉₉ = 20.5%
Worst	All three	WC, CL, EF, BI	EL = 757.31, VaR ₉₉ = 938.26, Δ VaR ₉₉ = 50.0%

Notes: Monetary amounts are in millions of dollars.

The baseline scenario assumes normal operations across all three systems. The moderate scenario captures a correlated disruption affecting Helionis and Oryn through shared communications, navigation, or logistics channels. The worst scenario reflects a severe event affecting all three systems through common transport, debris, or regulatory shocks. Each scenario is implemented using stress multipliers applied to baseline aggregate losses by hazard and solar

system. The resulting stressed portfolio outcomes are summarized in Table 4.3.

4.4 Threat table by risk level

Table 4.4 summarizes the main cross-system threats using ordinal risk levels from 1 to 5. The assessment is based on Table 4.1 and Section 4.2. Higher scores are assigned not only to threats with severe local impact, but also to those capable of generating multi-system disruption through shared transport, navigation, communications, or regulatory dependencies.

Table 4.4: Threat table by risk level across solar systems

Threat	Score	Higher-risk systems
Debris / navigation incidents	5	Helionis, Oryn
Supply chain / transport disruption	5	All systems, especially Helionis and Oryn
Communications / relay failure	4	Oryn, Helionis
Radiation / electromagnetic events	4	Bayesia, with possible spillover to Oryn
Operational expansion strain	3	Helionis, Bayesia
Environmental / regulatory change	3	All systems
Market competition	3	All systems
Equipment breakdown / maintenance strain	3	Helionis
Terrain hazards	3	Oryn

Notes: Risk levels are ordinal and range from 1 (very low) to 5 (very high), reflecting relative portfolio importance rather than probability alone.

4.5 Narrative of risk differences among solar systems

The three solar systems have distinct risk profiles: Helionis is mainly exposed to debris-related operational risk, Bayesia to radiation-driven stress, and Oryn Delta to navigation and infrastructure instability. These differences support system-specific underwriting adjustments.

- **Helionis Cluster:** Largest and most mature system, with higher Equipment Failure and Cargo Loss risk from older machinery and debris-heavy transit conditions.
- **Bayesia System:** More exposed to radiation and electromagnetic stress, leading to relatively higher Workers' Compensation and Business Interruption risk.
- **Oryn Delta:** Most unstable environment, with the highest Cargo Loss and Business Interruption risk due to debris, weak communications, and operational disruption.

4.6 ESG evaluation

ESG is used as an **underwriting-reference indicator** rather than a direct pricing variable. Bayesia and Helionis show relatively stronger ESG profiles, while Oryn ranks lower and therefore warrants more cautious underwriting. Table 4.5 reports the resulting scores. Further methodology is provided in Appendix H.

Table 4.5: Hazard-specific ESG scores by solar system

Solar system	WC	EF	CL	BI
Helionis	64.2	62.8	62.0	63.1
Bayesia	60.6	64.6	64.2	63.5
Oryn	49.4	53.8	52.9	52.5

5 Assumptions

- The severity distribution for each hazard is assumed to be common across solar systems, and all severity observations are assumed to be accurately recorded.
- Cosmic Quarry’s expansion is assumed to occur evenly over the next 10 years, with annual growth of 2.5% in Helionis and Bayesia, and 1.5% in Oryn, consistent with the stated long-term expansion plan.
- For the Workers’ Compensation transfer, the median claim amount in each occupation bucket is used to calculate the multiplier to reduce the influence of heavy tails.

6 Data and Data Limitations

For the Workers’ Compensation line, no directly comparable claims data are available for the new solar systems. Although the new portfolio provides detailed information on machinery, personnel, and operational exposures, there is no directly relevant historical loss experience under the new environmental conditions. Accordingly, an operational scaling ratio is used as a proxy to transfer exposure from the historical data. Since these new conditions are described mainly in qualitative terms, the pricing framework and risk assessment remain partly judgment-based and are subject to model uncertainty.

References

- [1] Sautner, Z., Van Lent, L., Vilkov, G. and Zhang, R., 2023. Firm-level climate change exposure. *The Journal of Finance*, 78(3), pp.1449-1498.

7 Disclaimer: AI Usage

AI tools were used in a limited supporting capacity during the preparation of this report:

- Language refinement and improvement of writing clarity
- Minor code editing and formatting adjustments
- Generation of preliminary brainstorming ideas during early project stages

All actuarial modelling, data analysis, and final interpretations were developed and verified by the authors. The authors retain full responsibility for the methodology and conclusions presented in this report.

A Benefit Structure Calibration

The main report summarises product design at a hazard-line level. This appendix provides additional rationale for the calibration of deductibles, limits and hazard-specific features.

A.1 Workers' Compensation

The deductible is calibrated at VaR_{50} severity to remove minor operational injury claims while preserving meaningful worker protection. The VaR_{95} limit controls extreme compensation events without imposing excessive retention on employees.

A.2 Equipment Failure

Equipment Failure losses are more heavy-tailed, particularly for *Quantum Bore*, which has substantially higher replacement value than other machinery. A VaR_{60} deductible removes smaller repair-type losses, while the VaR_{95} limit controls major breakdown claims. A specific VaR_{90} sublimit is imposed on *Quantum Bore* to manage concentrated high-value asset risk.

A.3 Cargo Loss

Cargo Loss adopts a VaR_{60} deductible to remove routine transit losses and a tighter VaR_{80} limit to better control large shipment claims. This tighter upper layer reflects the higher capital pressure created by cargo accumulation risk.

A.4 Business Interruption

Business Interruption losses arise from operational downtime and therefore accumulate over time rather than occurring as discrete damage events. Because the model simulates interruption losses directly in monetary form, a waiting period is approximated using a monetary deductible equivalent to two days of lost revenue.

The policy also introduces an annual aggregate cap calibrated at the VaR_{99} level of aggregate loss in order to control accumulation risk arising from prolonged disruptions.

A.5 Deductibles, Limits, BI Annual Caps and Loss Reduction

Table A.1: Deductibles, limits, BI annual caps and loss reduction by hazard and solar system

Solar System	Hazard	Deductible	Limit	BI Annual Cap	Loss Reduction
Bayesia System	BI	95,320,634	9,904,169	19,979,240,026	1.00
Bayesia System	CL	666,793	2,118,342	–	0.81
Bayesia System	WC	1,305	6,235	–	0.60
Bayesia System	EF	79,353	179,349	–	0.77
Helionis Cluster	BI	200,957,660	9,904,169	42,472,616,885	1.00
Helionis Cluster	CL	664,545	2,092,710	–	0.81
Helionis Cluster	EF	78,995	177,977	–	0.78
Helionis Cluster	WC	1,316	6,253	–	0.60
Oryn Delta	BI	59,354,583	9,904,169	13,503,901,033	0.99
Oryn Delta	CL	669,039	2,092,402	–	0.81
Oryn Delta	EF	80,471	182,808	–	0.77
Oryn Delta	WC	1,297	6,296	–	0.60

A.6 Equipment Failure Sublimit Calibration

Table A.2: Quantum Bore sublimit calibration

Equipment Type	VaR90 Severity	VaR95 Severity	Sublimit
Quantum Bore	215,888	263,641	179,772

A.7 Revenue Proxy Used for BI Waiting-Period Deductible

Hint on BI deductible calibration For Business Interruption, the waiting period is set at **2 days**. Based on the projected annual revenue in 2175 (this is calculated in the other Appendix), the total revenue across the three solar systems over two days is estimated as:

$$355,632,877 = \frac{\text{annual revenue in 2175}}{365} \times 2.$$

The BI deductible for each solar system is then calculated as:

$$\text{BI deductible for solar system } s = \text{proxy}_s \times 355,632,877,$$

where proxy_s is the equipment-based revenue allocation weight for solar system s .

Table A.3: Equipment-based solar system revenue proxy for BI deductible

Solar System	Revenue Proxy Weight
Bayesia System	0.2680
Helionis Cluster	0.5651
Oryn Delta	0.1669

B Hazard modeling

The following table presents the distributions we have fitted for the frequency and severity of the 4 hazards, including the predictors that are adopted:

Table B.1: Summary of final frequency and severity models

Hazard	Frequency Model	Predictors	Severity Model
Workers' Compensation	Poisson GLM	solar_system, occupation, accident_history_flag, psych_stress_index, safety_training_index	Spliced: Lognormal body + Weibull-type GPD tail
Cargo Loss	Quasi-Poisson GLM	route_risk, solar_radiation, de-bris_density, cargo_type, container_type, cargo_value, transit_duration, pilot_experience	Spliced: Lognormal body + Fréchet-type GPD tail
Business Interruption	Negative Binomial GLM	solar_system, production_load, energy_backup_score	Lognormal
Equipment Failure	Poisson GLM	equipment_type, solar_system, equipment_age, maintenance_int, usage_int	Quantum Bore: spliced Lognormal body + empirical bootstrap tail; other equipment: Lognormal

B.1 Frequency Modeling

B.1.1 General Framework

For each risk type, claim counts are modeled using a generalized linear model with a log link and an exposure offset. Let N_i denote the claim count for observation i , E_i denote the corresponding exposure, and X_i denote the vector of covariates. The generic specification is

$$\log(\mathbb{E}[N_i | X_i]) = \log(E_i) + \eta_i,$$

where η_i is the linear predictor.

Model family selection is based on the empirical variance-to-mean ratio, Pearson dispersion diagnostics, likelihood-based comparisons where applicable, out-of-sample validation, and actuarial interpretability.

B.1.2 Workers' Compensation Frequency

Workers' Compensation frequency is modeled using a Poisson GLM with log link and exposure offset. This choice is supported by the near-equidispersion of the claim counts after conditioning on the selected predictors.

Let N_i^{WC} denote the Workers' Compensation claim count. The final model is

$$N_i^{WC} \sim \text{Poisson}(\mu_i^{WC}),$$

with

$$\begin{aligned} \log(\mu_i^{WC}) = & \log(E_i) + \beta_0^{WC} + \beta_1^{WC} \text{accident_history_flag}_i \\ & + \sum_{s \in \mathcal{S}_{WC}^*} \beta_s^{WC, \text{solar}} \mathbf{1}\{\text{solar_system}_i = s\} + \sum_{o \in \mathcal{O}^*} \beta_o^{WC, \text{occ}} \mathbf{1}\{\text{occupation}_i = o\} \\ & + \beta_2^{WC} \text{psych_stress_index}_i + \beta_3^{WC} \text{safety_training_index}_i. \end{aligned}$$

Here, $\mathcal{S}_{WC}^*$ and $\mathcal{O}^*$ denote the non-reference levels of solar system and occupation, respectively.

The final specification was selected by combining domain knowledge, LASSO screening, and out-of-sample validation. In particular, psychological stress and safety training improved performance relative to the parsimonious benchmark.

B.1.3 Cargo Loss Frequency

Cargo Loss frequency is modelled using a Quasi-Poisson GLM. Although a Negative Binomial model improved in-sample likelihood, it overstated dispersion and performed poorly out-of-sample. By contrast, the Poisson mean structure calibrated well on the test set, so a Quasi-Poisson model was adopted to preserve the calibrated mean structure while allowing for overdispersion in inference.

Let N_i^{CL} denote the Cargo Loss claim count. The final model is

$$N_i^{CL} \sim \text{Quasi-Poisson}(\mu_i^{CL}, \phi_{CL}),$$

with

$$\begin{aligned} \log(\mu_i^{CL}) = & \log(E_i) + \beta_0^{CL} + \beta_1^{CL} \text{route_risk}_i + \beta_2^{CL} \text{solar_radiation}_i + \beta_3^{CL} \text{debris_density}_i \\ & + \sum_{c \in \mathcal{C}^*} \beta_c^{CL, \text{cargo}} \mathbf{1}\{\text{cargo_type}_i = c\} + \sum_{k \in \mathcal{K}^*} \beta_k^{CL, \text{container}} \mathbf{1}\{\text{container_type}_i = k\} \\ & + \beta_4^{CL} \text{cargo_value}_i + \beta_5^{CL} \text{transit_duration}_i + \beta_6^{CL} \text{pilot_experience}_i. \end{aligned}$$

Here, $\mathcal{C}^*$ and $\mathcal{K}^*$ denote the non-reference levels of cargo type and container type.

This model captures both environmental route hazards and shipment-specific operational features.

B.1.4 Business Interruption Frequency

Business Interruption claim counts exhibit material overdispersion relative to the Poisson benchmark. The baseline Poisson model shows a Pearson dispersion statistic substantially above one, and a formal likelihood ratio test strongly supports the Negative Binomial specification.

Let N_i^{BI} denote the Business Interruption claim count. The final model is

$$N_i^{BI} \sim \text{NB}(\mu_i^{BI}, \theta_{BI}),$$

with

$$\begin{aligned} \log(\mu_i^{BI}) = & \log(E_i) + \beta_0^{BI} + \sum_{s \in \mathcal{S}_{BI}^*} \beta_s^{BI, \text{solar}} \mathbf{1}\{S_i = s\} \\ & + \beta_1^{BI} \text{production_load}_i + \beta_2^{BI} \text{energy_backup_score}_i, \end{aligned}$$

Here, $\mathcal{S}_{BI}^*$ denotes the non-reference levels of solar system.

A richer specification with additional operational controls was considered, but the parsimonious model performed better out-of-sample and was therefore retained.

B.1.5 Equipment Failure Frequency

Equipment Failure frequency is modelled using a Poisson GLM. The claim counts are close to equidispersed both before and after conditioning on key predictors, and Pearson dispersion remains close to one across candidate specifications.

Let N_i^{EF} denote the Equipment Failure claim count. The final model is

$$N_i^{EF} \sim \text{Poisson}(\mu_i^{EF}),$$

with

$$\begin{aligned} \log(\mu_i^{EF}) = & \log(E_i) + \beta_0^{EF} + \sum_{e \in \mathcal{E}^*} \beta_e^{EF, \text{equip}} \mathbf{1}\{\text{equipment_type}_i = e\} \\ & + \sum_{s \in \mathcal{S}_{EF}^*} \beta_s^{EF, \text{solar}} \mathbf{1}\{\text{solar_system}_i = s\} \\ & + \beta_1^{EF} \text{equipment_age}_i + \beta_2^{EF} \text{maintenance_int}_i + \beta_3^{EF} \text{usage_int}_i. \end{aligned}$$

Here, $\mathcal{E}^*$ and $\mathcal{S}_{EF}^*$ denote the non-reference levels of equipment type and solar system.

The final specification was selected after comparing a parsimonious engineering model with an enriched model including maintenance and usage intensity. The enriched model achieved better out-of-sample performance and was therefore chosen.

B.2 Severity Modelling

B.2.1 General Framework

Severity modelling is conducted using positive claim amounts only. Candidate distributions are assessed using descriptive statistics, log-scale visual inspection, quantile diagnostics, tail concentration measures, and formal distributional comparisons. Where a single parametric distribution is insufficient to capture upper-tail behaviour, a spliced severity model is used.

B.2.2 Workers' Compensation Severity

Workers' Compensation severity exhibits an approximately Lognormal body with noticeable upper-tail deviation. A single standard parametric distribution is therefore insufficient for

tail-sensitive applications such as capital assessment.

The final severity model is a spliced Lognormal-GPD specification:

$$X^{WC} \sim \begin{cases} \text{Lognormal}(\mu_{WC}, \sigma_{WC}), & x \leq u_{WC}, \\ u_{WC} + \text{GPD}(\sigma_{WC}^{GPD}, \xi_{WC}), & x > u_{WC}. \end{cases}$$

Its cumulative distribution function is

$$F_{WC}(x) = \begin{cases} \phi_{WC} \frac{F_{LN}(x; \mu_{WC}, \sigma_{WC})}{F_{LN}(u_{WC}; \mu_{WC}, \sigma_{WC})}, & x \leq u_{WC}, \\ \phi_{WC} + (1 - \phi_{WC}) G_{GPD}(x - u_{WC}; \sigma_{WC}^{GPD}, \xi_{WC}), & x > u_{WC}, \end{cases}$$

where $\phi_{WC} = P(X^{WC} \leq u_{WC})$.

The fitted GPD shape parameter is expected to be negative, indicating a bounded Weibull-type tail rather than an unbounded Pareto tail.

B.2.3 Cargo Loss Severity

Cargo Loss severity exhibits a materially heavy upper tail. Tail concentration is substantial, and extreme value diagnostics support the use of a spliced model with a Generalized Pareto tail.

The final severity model is

$$X^{CL} \sim \begin{cases} \text{Lognormal}(\mu_{CL}, \sigma_{CL}), & x \leq u_{CL}, \\ u_{CL} + \text{GPD}(\sigma_{CL}^{GPD}, \xi_{CL}), & x > u_{CL}, \end{cases}$$

with $\xi_{CL} > 0$, indicating a Pareto-type heavy tail.

The spliced CDF is

$$F_{CL}(x) = \begin{cases} \phi_{CL}^{sev} \frac{F_{LN}(x; \mu_{CL}, \sigma_{CL})}{F_{LN}(u_{CL}; \mu_{CL}, \sigma_{CL})}, & x \leq u_{CL}, \\ \phi_{CL}^{sev} + (1 - \phi_{CL}^{sev}) G_{GPD}(x - u_{CL}; \sigma_{CL}^{GPD}, \xi_{CL}), & x > u_{CL}. \end{cases}$$

This model is preferred over single-distribution alternatives because it provides better fit to both the empirical survival function and upper-tail quantiles.

B.2.4 Business Interruption Severity

Business Interruption severity is adequately represented by a single Lognormal distribution. Although some diagnostics initially suggest upper-tail heaviness, the Hill plot does not reveal a stable Pareto tail index and the GPD stability plots do not support a valid spliced tail model. The Lognormal model therefore provides the best overall balance of body fit, tail fit, and parsimony.

The final model is

$$X^{BI} \sim \text{Lognormal}(\mu_{BI}, \sigma_{BI}),$$

with density

$$f_{BI}(x) = \frac{1}{x\sigma_{BI}\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu_{BI})^2}{2\sigma_{BI}^2}\right), \quad x > 0.$$

B.2.5 Equipment Failure Severity

Equipment Failure severity is modelled separately for Quantum Bore equipment and all other equipment types because the two groups exhibit materially different loss characteristics.

Quantum Bore Severity Quantum Bore claims exhibit a distinct catastrophic tail. A single full-support parametric model does not adequately describe both the body and the extreme observations. We therefore adopt a spliced specification with a truncated parametric body and an empirical bootstrap tail.

The body model is

$$X_{body}^{QB} \sim \text{Lognormal}(\mu_{QB}, \sigma_{QB}), \quad x \leq u_{QB},$$

while the tail is represented by the empirical distribution of the observed catastrophic losses

$$x_{QB,1}^{cat}, x_{QB,2}^{cat}, \dots, x_{QB,m}^{cat}.$$

The final CDF is

$$F_{QB}(x) = \begin{cases} \phi_{QB} \frac{F_{LN}(x; \mu_{QB}, \sigma_{QB})}{F_{LN}(u_{QB}; \mu_{QB}, \sigma_{QB})}, & x \leq u_{QB}, \\ \phi_{QB} + (1 - \phi_{QB}) G_{QB}^{cat}(x), & x > u_{QB}, \end{cases}$$

where G_{QB}^{cat} is the empirical CDF of the catastrophic sample.

This approach is preferred because the empirical tail is sparse and irregular, making direct

resampling of observed catastrophic losses more reliable than imposing a fully parametric tail.

Other Equipment Severity For non-Quantum Bore equipment, a single Lognormal distribution provides an adequate fit over the full support of the observed claims. Competing parametric specifications were considered, but the Lognormal model offered the best overall balance of goodness-of-fit and interpretability.

The final model is

$$X^{OE} \sim \text{Lognormal}(\mu_{OE}, \sigma_{OE}).$$

C Historical-to-New Data Transfer Methodology and Assumptions

C.1 Workers' Compensation Transfer Methodology

The Workers' Compensation (WC) analysis transfers the historical frequency–severity framework to the new solar-system portfolio using limited workforce and operational information. The transfer preserves the historical model structure while reconstructing the new portfolio on a comparable occupation and exposure basis.

The transfer has three steps. First, new personnel data are mapped into the historical occupation buckets and converted into effective exposure. Second, occupation-level frequency relativities are derived from the historical model and applied to the reconstructed exposure. Third, because workforce counts are unavailable by solar system, equipment counts and operating ratios are used to construct a solar-system activity proxy. Severity is transferred using a common spliced distribution with bucket-specific multipliers.

The transferred expected claim count for occupation bucket o is

$$\mu_o = E_o^{\text{eff}} \times \lambda_o^{\text{new}}, \quad \lambda_o^{\text{new}} = \lambda_{\text{base}} \times R_o \times F_{\text{solar,WC}},$$

where E_o^{eff} is effective exposure, R_o is the occupation relativity from the historical model, and $F_{\text{solar,WC}}$ is the solar adjustment factor.

C.1.1 Exposure reconstruction from new personnel data

The new portfolio provides role-level employee counts split into full-time and contract staff. These are converted into effective exposure using historical average exposure by employment type.

Detailed roles are mapped into four WC occupation buckets:

- Office/Admin,
- Technical,
- Field Operations,
- Heavy Operations.

Let $\bar{e}_{\text{FT}}$ and $\bar{e}_{\text{CT}}$ be historical mean exposure for full-time and contract workers. Effective exposure for bucket o is

$$E_o^{\text{eff}} = N_{o,\text{FT}}\bar{e}_{\text{FT}} + N_{o,\text{CT}}\bar{e}_{\text{CT}},$$

where $N_{o,\text{FT}}$ and $N_{o,\text{CT}}$ are the corresponding worker counts.

C.1.2 Occupation relativities from the historical frequency model

The historical WC frequency model is used to recover fitted claim rates. For each historical observation i ,

$$\hat{\mu}_i = \widehat{\mathbb{E}}[N_i \mid X_i], \quad \hat{\lambda}_i = \frac{\hat{\mu}_i}{E_i}.$$

Historical occupations are mapped into the same four buckets, and the fitted claim rate for bucket o is

$$\hat{\lambda}_o = \frac{\sum_{i \in o} \hat{\mu}_i}{\sum_{i \in o} E_i}.$$

The Technical bucket is used as the reference:

$$\lambda_{\text{base}} = \hat{\lambda}_{\text{Technical}}.$$

The occupation relativity is

$$R_o = \frac{\hat{\lambda}_o}{\lambda_{\text{base}}}.$$

C.1.3 Solar-system matching using operational proxies

Because workforce counts are not available by solar system, WC activity is allocated using equipment counts and operating ratios.

For equipment type e in system s ,

$$A_{s,e} = C_{s,e} \times p_{s,e},$$

where $C_{s,e}$ is equipment count and $p_{s,e}$ is the proportion in operation. Aggregating across equipment types gives

$$W_s = \sum_e A_{s,e},$$

and the corresponding system weight is

$$w_s = \frac{W_s}{\sum_s W_s}.$$

Qualitative system risk factors F_s are then assigned and combined with the proxy weights to form

$$F_{\text{solar,WC}} = \sum_s w_s F_s.$$

Hence the transferred frequency rate becomes

$$\lambda_o^{\text{new}} = \lambda_{\text{base}} \times R_o \times F_{\text{solar,WC}},$$

and the expected claim count is

$$\mu_o = E_o^{\text{eff}} \times \lambda_o^{\text{new}}.$$

C.1.4 Severity transfer

Severity is transferred using a common spliced distribution with a Lognormal body and a Generalised Pareto tail. Let u be the splice threshold. The baseline severity distribution is

$$F(x) = \begin{cases} \phi_u \frac{F_{\text{LN}}(x)}{F_{\text{LN}}(u)}, & x \leq u, \\ \phi_u + (1 - \phi_u) F_{\text{GPD}}(x - u), & x > u. \end{cases}$$

Bucket-specific differences are introduced using historical median claim size. If m_o is the

median severity for bucket o , the multiplier is

$$M_o^{\text{sev}} = \frac{m_o}{m_{\text{Technical}}}.$$

In simulation, baseline severities are sampled from the spliced distribution and scaled by M_o^{sev} .

C.1.5 Aggregate loss simulation

For each occupation bucket o ,

$$N_o \sim \text{Poisson}(\mu_o).$$

This mean is allocated across solar systems using the system weights:

$$\mu_{so} = \mu_o w_s.$$

For each bucket-system cell,

$$N_{so} \sim \text{Poisson}(\mu_{so}), \quad S_{so} = \sum_{k=1}^{N_{so}} X_{so,k},$$

where $X_{so,k}$ are simulated severities after applying the bucket-specific multiplier. Total annual WC loss is

$$S_{\text{WC}} = \sum_s \sum_o S_{so}.$$

Monte Carlo simulation is repeated 10,000 times. Summary measures are reported at four levels:

1. total portfolio,
2. by occupation bucket,
3. by solar system,
4. by occupation bucket $\times$ solar system.

Table C.1: WC transfer coefficients and multipliers

Component	Symbol	Formula	Purpose
Mean full-time exposure	$\bar{e}_{\text{FT}}$	$\text{mean}(E_i \mid \text{full-time})$	Converts new full-time headcount into historical-equivalent exposure units.
Mean contract exposure	$\bar{e}_{\text{CT}}$	$\text{mean}(E_i \mid \text{contract})$	Converts new contract headcount into historical-equivalent exposure units.
Effective exposure	E_o^{eff}	$N_{o,\text{FT}}\bar{e}_{\text{FT}} + N_{o,\text{CT}}\bar{e}_{\text{CT}}$	Reconstructs WC exposure for each occupation bucket.
Base claim rate	λ_{base}	$\hat{\lambda}_{\text{Technical}}$	Reference fitted claim rate from the historical model.
Occupation relativity	R_o	$\hat{\lambda}_o / \lambda_{\text{base}}$	Captures bucket-level frequency differences.
Active units	$A_{s,e}$	$C_{s,e} \times p_{s,e}$	Measures effective operating equipment by system and type.
System scale	W_s	$\sum_e A_{s,e}$	Proxy for WC operational activity in each solar system.
System weight	w_s	$W_s / \sum_s W_s$	Allocates portfolio activity across solar systems.
Solar risk factor	F_s	judgement-based input	Represents relative operating risk by solar system.
Portfolio solar adjustment	$F_{\text{solar},\text{WC}}$	$\sum_s w_s F_s$	Produces a single frequency adjustment for the new portfolio mix.
Transferred claim rate	λ_o^{new}	$\lambda_{\text{base}} R_o F_{\text{solar},\text{WC}}$	Transfers frequency assumptions from the historical model to the new portfolio.
Expected claim count	μ_o	$E_o^{\text{eff}} \lambda_o^{\text{new}}$	Frequency input for aggregate loss simulation.
Severity multiplier	M_o^{sev}	$m_o / m_{\text{Technical}}$	Adjusts baseline claim severity by occupation bucket.

C.2 Equipment Failure Transfer Methodology

The Equipment Failure (EF) analysis transfers the historical frequency–severity framework to the new portfolio at the solar system $\times$ equipment type level. The main issue is that the new solar systems are not valid factor levels in the historical GLM. The transfer therefore separates model-based prediction from external adjustment.

The approach has four steps. First, each new stratum is mapped to the historical model structure using the same equipment types and a baseline historical solar-system level. Second, exposure is reconstructed from machine counts and utilisation ratios. Third, system

differences not captured by the historical GLM are introduced through an external equipment-level multiplier. Fourth, the adjusted frequency model is combined with historical severity distributions in Monte Carlo simulation.

For stratum s, e ,

$$\mu_{se} = \lambda_{se}^{\text{adj}}, \quad \lambda_{se}^{\text{adj}} = \lambda_{se}^{\text{GLM}} \times M_{se}^{\text{eq}},$$

where $\lambda_{se}^{\text{GLM}}$ is the historical-model prediction and M_{se}^{eq} is the residual equipment-level multiplier.

C.2.1 Prediction strata and portfolio mapping

The EF transfer is performed at the

$$(\text{solar system, equipment type})$$

level. Raw equipment names in the new portfolio are first mapped to the historical factor labels used in the fitted GLM.

C.2.2 Exposure reconstruction from inventory and operation ratios

For each stratum, exposure is reconstructed from machine counts and operation ratios. Let

$$N_{se} = \text{machine count for system } s \text{ and equipment } e,$$

and

$$\rho_{se} = \text{average exposure ratio per machine.}$$

Exposure is then

$$E_{se} = N_{se} \rho_{se}.$$

C.2.3 Deriving equipment age from service-year distributions

The new data provide service-year bands rather than exact ages. Expected age is approximated using band midpoints.

Let c_{seb} be the number of machines in band b , and let m_b be the band midpoint:

$$< 5 \mapsto 2.5, \quad 5-9 \mapsto 7, \quad 10-14 \mapsto 12, \quad 15-19 \mapsto 17, \quad 20+ \mapsto 22.$$

Then

$$A_{se} = \frac{\sum_b c_{seb} m_b}{\sum_b c_{seb}}.$$

C.2.4 Construction of historical-model inputs

The historical EF model requires equipment type, solar system, equipment age, maintenance interval, usage intensity, and exposure. Since the new solar systems are not valid factor levels, all new strata are predicted using the baseline historical level

$$\text{solar_system_model} = \text{Helionis Cluster}.$$

Thus,

$$\lambda_{se}^{\text{GLM}} = \widehat{\mathbb{E}}[N_{se} \mid \text{equipment type} = e, \text{solar system} = \text{Helionis}, A_{se}, \text{maintenance}_{se}, \text{usage}_e, E_{se}].$$

Usage intensity in the new data is approximated by the historical mean for equipment type e :

$$U_e = \text{mean}(\text{usage intensity} \mid \text{equipment type} = e).$$

If unavailable, the overall historical mean is used.

C.2.5 Residual equipment-level risk multipliers

Residual cross-system differences are introduced outside the GLM using qualitative equipment-level risk scores. Let R_{se} be the raw score for system s and equipment e . Helionis Cluster is the baseline, so

$$M_{se}^{\text{eq}} = \frac{R_{se}}{R_{\text{Helionis},e}}.$$

Hence,

$$M_{\text{Helionis},e}^{\text{eq}} = 1,$$

and the adjusted expected claim count is

$$\lambda_{se}^{\text{adj}} = \lambda_{se}^{\text{GLM}} \times M_{se}^{\text{eq}}.$$

C.2.6 Severity transfer

Severity follows the historical EF severity structure with two groups:

-
- **Group A:** Quantum Bore,
 - **Group B:** all other equipment types.

For Quantum Bore,

$$X \sim F_A,$$

and for all other equipment,

$$X \sim F_B.$$

C.2.7 Aggregate loss simulation

For each stratum (s, e) ,

$$N_{se} \sim \text{Poisson}(\lambda_{se}^{\text{adj}}).$$

If $N_{se} > 0$,

$$S_{se} = \sum_{k=1}^{N_{se}} X_{sek}.$$

Total annual EF loss is

$$S_{\text{EF}} = \sum_s \sum_e S_{se}.$$

Monte Carlo simulation is repeated 10,000 times. Summary measures are reported at three levels:

1. total portfolio,
2. by solar system,
3. by solar system $\times$ equipment type.

Table C.2: EF severity and simulation structure

Component	Symbol	Formula	Purpose
Severity model for Quantum Bore	F_A	historical Group A severity	Preserves the distinct severity behaviour of Quantum Bore claims.
Severity model for other equipment	F_B	historical Group B severity	Provides baseline severity for all non-Quantum Bore equipment.
Simulated claim count	N_{se}	$\text{Poisson}(\lambda_{se}^{\text{adj}})$	Generates annual claim frequency for each solar system $\times$ equipment type stratum.
Aggregate stratum loss	S_{se}	$\sum_{k=1}^{N_{se}} X_{sek}$	Aggregates simulated claim severities within each stratum.
Total EF annual loss	S_{EF}	$\sum_s \sum_e S_{se}$	Produces total simulated annual portfolio loss.
Monte Carlo runs	–	10,000 simulations	Generates the empirical annual loss distribution for risk and pricing metrics.

Table C.3: EF frequency transfer inputs and adjustments

Component (Symbol)	Formula	Purpose
Machine count (N_{se})	inventory input	Number of machines in solar system s for equipment type e .
Exposure ratio per machine (ρ_{se})	operational input	Measures the proportion of time each machine is in operation.
Exposure (E_{se})	$N_{se}\rho_{se}$	Reconstructs effective machine exposure for the historical model.
Age-band midpoint (m_b)	fixed midpoint by band	Converts grouped service-year information into approximate numerical age.
Expected equipment age (A_{se})	$\sum_b c_{seb}m_b / \sum_b c_{seb}$	Produces a continuous age input for GLM prediction.
Maintenance interval (Maint $_{se}$)	operational input	Captures maintenance differences across systems and equipment types.
Historical usage intensity (U_e)	mean(usage intensity e)	Supplies model-compatible usage input when the new data are not directly comparable.
Baseline prediction system	Helionis Cluster	Historical solar-system level used inside <code>predict()</code> for all new rows.
GLM-predicted claim count ($\lambda_{se}^{\text{GLM}}$)	$\hat{\mathbb{E}}[N_{se} \mid \text{GLM inputs}]$	Historical-model frequency prediction for each new stratum.
Raw risk score (R_{se})	qualitative risk table	Captures residual equipment-system differences outside the historical GLM.
Equipment residual multiplier (M_{se}^{eq})	$R_{se}/R_{\text{Helionis},e}$	Adjusts GLM predictions for residual solar-system risk by equipment type.
Adjusted expected claim count ($\lambda_{se}^{\text{adj}}$)	$\lambda_{se}^{\text{GLM}}M_{se}^{\text{eq}}$	Final frequency input used in Monte Carlo simulation.

C.3 Cargo Loss Transfer Methodology

The Cargo Loss (CL) analysis transfers the historical frequency–severity framework to the new portfolio at the solar system $\times$ container type level. Because the new portfolio does not contain shipment-level claim data, the transfer uses historical prototype matching together

with exposure calibration back to the historical model scale.

The approach has four steps. First, each new stratum is defined by solar system and container type. Second, exposure is reconstructed from container quantity and capacity, then calibrated to the historical exposure basis. Third, the historical frequency model is transferred using container-type prototypes with new-system environmental overrides. Fourth, severity is modelled by container type and scaled to the transferred cargo-value level.

For stratum s, c ,

$$\mu_{sc} = \lambda_{sc}^{\text{GLM}},$$

where $\lambda_{sc}^{\text{GLM}}$ is the expected claim count predicted from reconstructed new-data inputs.

C.3.1 Prediction strata and prototype matching

The new cargo portfolio is represented at the

(solar system, container type)

level. Raw container names are first mapped to the historical container-type labels used in the fitted cargo model.

Because shipment-level covariates are unavailable, a historical prototype is built for each container type. For each c , the prototype contains:

- modal cargo type,
- median cargo value,
- median transit duration,
- median pilot experience,
- historical median solar radiation,
- historical median debris density.

Under Route B transfer, each new container type inherits its historical prototype profile, with solar-system-specific environmental variables replaced by the new system values.

C.3.2 Exposure reconstruction and calibration

A capacity proxy is first defined as

$$K_{sc} = Q_{sc} \times V_c,$$

where Q_{sc} is the number of containers and V_c is the maximum volume per container type.

This is then converted to the historical exposure scale using

$$\kappa_{\text{hist}} = \text{median} \left(\frac{\text{weight}}{\text{exposure}} \right),$$

so that

$$E_{sc} = \frac{K_{sc}}{\kappa_{\text{hist}}}.$$

C.3.3 System-level environmental overrides

A system-environment table is constructed for each solar system, containing:

- route risk,
- solar radiation,
- debris density,
- transit multiplier,
- pilot multiplier.

For stratum (s, c) ,

cargo type_{sc} = historical prototype cargo type for c ,

cargo value_{sc} = historical prototype cargo value for c ,

transit duration_{sc} = historical prototype transit duration for $c \times$ transit multiplier_s,

pilot experience_{sc} = historical prototype pilot experience for $c \times$ pilot multiplier_s.

Route risk, solar radiation, and debris density are taken directly from the new system table.

C.3.4 Historical frequency prediction

The historical cargo frequency model uses:

route risk, solar radiation, debris density, cargo type, container type, cargo value, transit duration, pilot

together with exposure if included in the fitted specification.

After reconstructing the inputs, expected claim count is predicted as

$$\lambda_{sc}^{\text{GLM}} = \widehat{\mathbb{E}}[N_{sc} \mid X_{sc}],$$

where X_{sc} is the transferred predictor vector.

C.3.5 Dispersion treatment for frequency simulation

The fitted cargo frequency model is Quasi-Poisson, so a simulation distribution must be imposed. The model dispersion parameter ϕ is matched to a Negative Binomial approximation.

For Quasi-Poisson,

$$\text{Var}(N_{sc}) = \phi \mu_{sc}.$$

For Negative Binomial with mean μ_{sc} and size k_{sc} ,

$$\text{Var}(N_{sc}) = \mu_{sc} + \frac{\mu_{sc}^2}{k_{sc}}.$$

Matching the variances gives

$$k_{sc} = \frac{\mu_{sc}}{\phi - 1}, \quad \text{when } \phi > 1.$$

Hence,

$$N_{sc} \sim \begin{cases} \text{Negative Binomial}(\mu_{sc}, k_{sc}), & \phi > 1, \\ \text{Poisson}(\mu_{sc}), & \phi \leq 1. \end{cases}$$

C.3.6 Severity transfer by container type

Severity is modelled separately for each historical container type. For each type c , positive claim amounts are fitted using:

- a Lognormal body below threshold u_c , and

- a Generalised Pareto tail above u_c .

If the tail sample is too small, a full Lognormal distribution is used.

Each container type c is assigned a fitted severity object F_c . Simulated severity is then scaled by cargo value:

$$X_{sc}^{\text{new}} = X_c^{\text{base}} \left(\frac{CV_{sc}}{CV_c^{\text{ref}}} \right)^\alpha,$$

where $\alpha = 0.70$.

C.3.7 Aggregate loss simulation

For each stratum (s, c) , annual claim frequency is simulated from the NB or Poisson model above. Conditional on N_{sc} , severities are sampled from the fitted container-type severity model and scaled by cargo value:

$$S_{sc} = \sum_{k=1}^{N_{sc}} X_{sck}.$$

Total annual Cargo Loss is

$$S_{\text{CL}} = \sum_s \sum_c S_{sc}.$$

Monte Carlo simulation is repeated 10,000 times. Summary measures are reported at three levels:

1. total portfolio,
2. by solar system,
3. by solar system $\times$ container type.

Table C.4: Cargo frequency transfer inputs and adjustments

Component (Symbol)	Formula	Purpose
Container quantity (Q_{sc})	inventory input	Number of containers in solar system s for container type c .
Maximum volume (V_c)	inventory input	Container-specific capacity used to build the shipment exposure proxy.
Capacity proxy (K_{sc})	$Q_{sc} V_c$	Measures shipment scale before calibration back to the historical exposure basis.
Historical exposure calibration (κ_{hist})	median(weight/exposure)	Converts the new capacity proxy into the historical model exposure scale.
Transferred exposure (E_{sc})	$K_{sc}/\kappa_{\text{hist}}$	Reconstructs model-compatible exposure for each stratum.
Historical prototype profile	by container type	Supplies representative historical cargo type, value, transit duration, and pilot experience.
Route risk	system input	Captures route-level operating risk in the new solar system.
Solar radiation	system input	Overrides the historical environmental level with new system conditions.
Debris density	system input	Captures the new system's external physical hazard environment.
Transit-duration adjustment	prototype duration $\times$ transit multiplier _{s}	Adapts historical prototype journey length to the new solar system.
Pilot-experience adjustment	prototype experience $\times$ pilot multiplier _{s}	Adjusts the historical prototype pilot profile for the new system.
GLM-predicted claim count ($\lambda_{sc}^{\text{GLM}}$)	$\hat{\mathbb{E}}[N_{sc} X_{sc}]$	Historical-model expected claim count for each new cargo stratum.
Dispersion parameter (ϕ)	from quasi-Poisson fit	Captures extra-Poisson variability in the cargo frequency model.
NB size parameter (k_{sc})	$\mu_{sc}/(\phi - 1)$	Provides a moment-matched simulation distribution when $\phi > 1$.

Table C.5: Cargo severity transfer and simulation structure

Component (Symbol)	Formula	Purpose
Severity model by container type (F_c)	Lognormal body + GPD tail	Preserves historical claim-size behaviour separately for each container type.
Fallback severity model	Lognormal only	Used when tail observations are insufficient for stable GPD fitting.
Reference cargo value (CV_c^{ref})	historical median by container type	Anchors the severity model to the historical insured-value level.
New cargo value (CV_{sc})	transferred prototype value	Represents the cargo-value level used for severity scaling in the new stratum.
Cargo-value scaling	$(CV_{sc}/CV_c^{\text{ref}})^\alpha$	Adjusts claim size for differences between historical and new cargo-value levels.
Scaling elasticity (α)	0.70	Controls the sensitivity of simulated severity to cargo-value differences.
Simulated claim count (N_{sc})	NB or Poisson	Generates annual claim frequency for each stratum.
Aggregate stratum loss (S_{sc})	$\sum_{k=1}^{N_{sc}} X_{sck}$	Aggregates simulated severities within each cargo stratum.
Total cargo annual loss (S_{CL})	$\sum_s \sum_c S_{sc}$	Produces total simulated annual portfolio loss.
Monte Carlo runs	10,000 simulations	Generates the empirical annual loss distribution for pricing and risk metrics.

C.4 Business Interruption Transfer Methodology

The Business Interruption (BI) analysis transfers the historical frequency–severity framework to the new portfolio using a proxy-based approach. Unlike the other hazard lines, the new data do not contain detailed operating records that can be matched directly to the historical BI model. The transfer is therefore performed at the solar-system level using proxy mapping, exposure scaling, and scenario-based frequency simulation.

The approach has four steps. First, each new solar system is mapped to one or more historical proxy systems. Second, total exposure is reconstructed from equipment counts using Helionis

Cluster as the calibration anchor. Third, expected BI frequency is predicted on weighted proxy rows using the historical Negative Binomial model. Fourth, aggregate BI loss is simulated under alternative dispersion scenarios using a Lognormal severity model fitted to historical claims.

For new solar system s ,

$$\mu_s = \sum_{j \in \mathcal{P}(s)} \mu_{sj}^{\text{proxy}},$$

where $\mathcal{P}(s)$ is the set of historical proxy systems and μ_{sj}^{proxy} is the predicted count for proxy row j .

C.4.1 Proxy mapping from new systems to historical systems

Each new solar system is linked to one or more historical systems already included in the fitted BI model. For system s , the proxy table specifies:

- historical proxy systems,
- a weight w_{sj} for each proxy system j ,
- qualitative justification.

These weights satisfy

$$\sum_{j \in \mathcal{P}(s)} w_{sj} = 1.$$

Helionis Cluster is treated as a direct match, while Bayesia System and Oryn Delta are represented as weighted combinations of historical systems.

C.4.2 Exposure reconstruction using equipment-scale calibration

Because direct BI exposure is unavailable, system-level exposure is reconstructed using total equipment count. Let M_s be the total equipment count for new solar system s . Using Helionis Cluster as the anchor,

$$E_{\text{Hel}}^{\text{hist}} = \text{historical total BI exposure for Helionis Cluster},$$

and

$$M_{\text{Hel}} = \text{new equipment total for Helionis Cluster}.$$

The assumed total exposure for system s is

$$E_s^{\text{assumed}} = E_{\text{Hel}}^{\text{hist}} \times \frac{M_s}{M_{\text{Hel}}}.$$

This is then split across proxy rows:

$$E_{sj}^{\text{assumed}} = E_s^{\text{assumed}} \times w_{sj}.$$

C.4.3 Historical proxy statistics

For each historical proxy system j , the following summaries are extracted:

- mean production load,
- modal energy backup score,
- total exposure,
- implied historical claim rate.

These are used as assumed model inputs:

$$\text{production load}_{sj} = \overline{\text{production load}_j},$$

$$\text{energy backup score}_{sj} = \text{mode}(\text{energy backup score}_j).$$

C.4.4 Historical frequency prediction on proxy rows

The fitted BI frequency model uses

solar system, production load, energy backup score, exposure.

For each proxy row (s, j) ,

$$\mu_{sj}^{\text{proxy}} = \widehat{\mathbb{E}}[N_{sj} \mid \text{solar system} = j, \text{production load}_{sj}, \text{energy backup score}_{sj}, E_{sj}^{\text{assumed}}].$$

The total expected claim count is

$$\mu_s = \sum_{j \in \mathcal{P}(s)} \mu_{sj}^{\text{proxy}},$$

and the implied claim rate is

$$\lambda_s^{\text{implied}} = \frac{\mu_s}{E_s^{\text{assumed}}}.$$

C.4.5 Severity transfer

BI severity is transferred using a single Lognormal distribution fitted to historical positive claims:

$$\log Y \sim N(\mu_{\text{BI}}, \sigma_{\text{BI}}^2),$$

where

$$\mu_{\text{BI}} = \text{mean}(\log Y), \quad \sigma_{\text{BI}} = \text{sd}(\log Y).$$

C.4.6 Dispersion scenarios for frequency simulation

Because BI frequency is transferred through proxy rows rather than direct record-level matching, several dispersion settings are tested.

Three scenarios are used:

- $\theta = 1$,
- $\theta = 5$,
- $\theta = \theta_{\text{fitted}}$.

For each solar system s ,

$$N_s \sim \text{Negative Binomial}(\mu_s, \theta),$$

where μ_s is the transferred expected claim count and θ is the scenario-specific dispersion parameter.

C.4.7 Aggregate loss simulation

For each solar system s ,

$$S_s = \sum_{k=1}^{N_s} X_{sk}, \quad X_{sk} \sim \text{Lognormal}(\mu_{\text{BI}}, \sigma_{\text{BI}}).$$

Total annual BI loss is

$$S_{\text{BI}} = \sum_s S_s.$$

Monte Carlo simulation is repeated 10,000 times for each theta scenario. Summary measures are reported at two levels:

1. total portfolio,
2. by solar system.

Table C.6: BI frequency transfer inputs and proxy structure

Component (Symbol)	Formula	Purpose
New solar system (s)	portfolio input	Target solar system for BI transfer pricing.
Historical proxy system (j)	proxy mapping input	Historical system used to represent the new solar system in the fitted BI model.
Proxy weight (w_{sj})	mapping assumption	Allocates each new solar system across one or more historical proxy systems.
Equipment total (M_s)	operational proxy	Represents operational scale when direct BI exposure is unavailable.
Helionis historical exposure ($E_{\text{Hel}}^{\text{hist}}$)	historical total	Calibration anchor for reconstructing new-system exposure.
Assumed total exposure (E_s^{assumed})	$E_{\text{Hel}}^{\text{hist}}(M_s/M_{\text{Hel}})$	Scales BI exposure for each new solar system using equipment counts.
Proxy-row exposure (E_{sj}^{assumed})	$E_s^{\text{assumed}} w_{sj}$	Splits new-system exposure across weighted proxy rows.
Assumed production load	historical mean by proxy system	Supplies model-compatible operating intensity for each proxy row.
Assumed energy backup score	historical mode by proxy system	Supplies the representative resilience category for each proxy row.
Proxy-row predicted claim count (μ_{sj}^{proxy})	$\widehat{\mathbb{E}}[N_{sj} \mid \text{proxy inputs}]$	Historical-model expected claim count for each weighted proxy row.
Transferred expected claim count (μ_s)	$\sum_{j \in \mathcal{P}(s)} \mu_{sj}^{\text{proxy}}$	Final expected annual BI claim count for the new solar system.
Implied claim rate ($\lambda_s^{\text{implied}}$)	$\mu_s / E_s^{\text{assumed}}$	Converts predicted frequency into a claim rate per unit assumed exposure.

Table C.7: BI severity transfer and simulation settings

Component (Symbol)	Formula	Purpose
Severity model	Lognormal	Transferred BI severity distribution fitted to historical positive claim amounts.
Lognormal meanlog (μ_{BI})	$\text{mean}(\log Y)$	Central level of historical BI severity on the log scale.
Lognormal sdlog (σ_{BI})	$\text{sd}(\log Y)$	Dispersion of historical BI severity on the log scale.
Dispersion scenario (θ)	$\{1, 5, \theta_{\text{fitted}}\}$	Sensitivity settings for BI frequency dispersion.
Simulated claim count (N_s)	$\text{NB}(\mu_s, \theta)$	Generates annual BI frequency for each solar system.
Aggregate system loss (S_s)	$\sum_{k=1}^{N_s} X_{sk}$	Aggregates simulated BI severities within each solar system.
Total BI annual loss (S_{BI})	$\sum_s S_s$	Produces total simulated annual BI portfolio loss.
Monte Carlo runs	10,000 per scenario	Generates the empirical annual loss distribution.

C.5 Model limitations

Several limitations should be noted in the transfer framework. First, all four hazard lines rely on historical-model transfer rather than direct re-estimation on new-system claims, so the results are subject to model risk and possible non-stationarity between the historical and new portfolios. Second, the new portfolio data are incomplete, which requires the use of exposure reconstruction, prototype matching, proxy mapping, and judgement-based multipliers. These assumptions introduce approximation error and mean that the outputs should be interpreted as pricing indications under limited information rather than fully credible experience-based premiums.

At the hazard level, the main limitations differ by line. For Workers' Compensation, the reconstructed effective exposure and role-to-bucket mapping may not fully capture the underlying labour-risk structure, and the use of Poisson simulation may understate count volatility. For Equipment Failure, new solar systems are represented through a historical

baseline system with external equipment-level multipliers, which may miss interaction effects between system conditions and equipment characteristics. For Cargo Loss, prototype matching compresses shipment-level heterogeneity, while exposure calibration and cargo-value scaling are approximate rather than directly observed. For Business Interruption, the transfer is the most assumption-driven, relying on proxy-system weights, equipment-based exposure scaling, and common severity assumptions in the absence of detailed new operational data.

These limitations were partially addressed through sensitivity analysis, conservative proxy choices, and transparent documentation of assumptions; however, the transferred results remain dependent on the stability of historical relationships and the reasonableness of the proxy structure.

D Scenario stress methodology

D.1 Technical implementation of stress overlays

The scenario testing in Section 3.2 is implemented by applying explicit stress multipliers directly to baseline aggregate losses, rather than re-parameterising the underlying frequency and severity models. This provides a practical top-down approach for assessing adverse portfolio outcomes while remaining consistent with the fitted loss modelling framework.

For hazard coverage h , solar system s , and stress scenario k , let $L_{h,s}$ denote the baseline annual aggregate loss. Under scenario k , the stressed aggregate loss is defined as

$$L_{h,s}^{(k)} = m_{h,s,k}^{\text{loss}} L_{h,s},$$

where $m_{h,s,k}^{\text{loss}}$ is the scenario-specific loss multiplier.

This formulation is intended to capture, in reduced form, the combined effect of higher claim frequency, greater claim severity, and stronger cross-risk dependence under adverse conditions, without separately stressing each component of the underlying loss process.

Portfolio aggregation. Total portfolio loss under scenario k is obtained by summing stressed losses across all relevant hazard coverages and solar systems:

$$L^{(k)} = \sum_h \sum_s L_{h,s}^{(k)}.$$

Required capital. Required capital under scenario k is defined as

$$\text{Required Capital}^{(k)} = \text{VaR}_{99}^{(k)} - \mathbb{E}[L^{(k)}],$$

where $L^{(k)}$ denotes stressed aggregate portfolio loss.

Table D.1: Stress overlay notation

Symbol	Description
$L_{h,s}$	Baseline annual aggregate loss for hazard h in solar system s
$m_{h,s,k}^{\text{loss}}$	Scenario- k multiplier applied to aggregate loss
$L_{h,s}^{(k)}$	Stressed aggregate loss for hazard h in solar system s under scenario k
$L^{(k)}$	Total stressed portfolio loss under scenario k

D.2 Illustrative stress overlay inputs

Table D.2 reports illustrative stress multipliers for the scenarios considered in Section 3.2. These are intended as transparent starting assumptions and may be refined after reviewing the resulting stressed loss distributions, capital requirements, and pricing implications.

Table D.2: Illustrative stress overlay inputs by scenario

Scenario	Systems	Hazard	Multipliers	Rationale
Best	All three	WC	1.00	Attritional baseline
	All three	CL	1.00	Attritional baseline
	All three	EF	1.00	Attritional baseline
	All three	BI	1.00	Attritional baseline
Moderate	Helionis + Oryn	WC	1.00	No material direct workforce stress assumed
	Helionis + Oryn	CL	1.30	Two-system navigation / logistics disruption
	Helionis + Oryn	EF	1.25	Equipment and systems disruption under stressed operations
	Helionis + Oryn	BI	1.35	Contained but meaningful operating interruption
Worst	All three	WC	1.10	Secondary workforce stress under firm-wide disruption
	All three	CL	1.50	Network-wide logistics and debris shock
	All three	EF	1.35	Firm-wide equipment and systems stress
	All three	BI	1.55	Prolonged interruption under severe correlated shock

Notes: WC = Workers’ Compensation; CL = Cargo Loss; EF = Equipment Failure; BI = Business Interruption. “Loss mult.” denotes a multiplier applied directly to baseline aggregate loss. The values above are illustrative starting assumptions and may be recalibrated after reviewing stressed simulation outputs.

E Additional correlated risk scenario details

This appendix provides further detail on the correlated risk scenarios used in the risk assessment. Since Helionis Cluster, Bayesia System, and Oryn Delta are distinct stellar systems, multi-system correlation is not interpreted as a single local stellar-weather event affecting all systems simultaneously. Instead, correlated scenarios are defined through either common interstellar shocks or shared operational dependencies, such as communications, transport, navigation, and regulatory channels.

Scenario 1: Deep-space radiation transient affecting Bayesia and Oryn

A high-energy deep-space radiation transient is a plausible joint shock for Bayesia and Oryn. Bayesia is already exposed to temporary spikes of electromagnetic and particle radiation under binary-star orbital alignments, and its mineable planet has a thin magnetosphere and

elevated ambient radiation exposure. Oryn, while currently centred on a relatively mild dwarf star, is described as vulnerable to flare uncertainty in similar systems and already operates with fragile long-range communications and deep-extraction infrastructure. A deep-space radiation event passing through the operating corridor could therefore simultaneously impair shielding, control electronics, relay reliability, and remote extraction systems in both solar systems.

This scenario is expected to affect Equipment Failure, Cargo, and Business Interruption most strongly. Workers' Compensation exposure is likely to be secondary, arising only if operational disruption or hazardous repair activity increases staff risk.

Scenario 2: Interstellar relay and communications backbone failure affecting Bayesia and Oryn

A second correlated scenario arises from a failure in the shared communications and relay architecture linking off-world operations. Bayesia relies on orbital stations and communication relays to support standardised supply routes, while Oryn depends on amplified beacons and mobile relay drones because long-range communications remain inconsistent. A disruption to a shared relay corridor, backbone routing node, or interstellar communications chain could therefore simultaneously impair operations in both systems.

This scenario would primarily produce Cargo delay, repair delay, and Business Interruption losses, with secondary spillovers into Equipment Failure if communication loss prevents remote diagnostics, navigation updates, or timely maintenance. Workers' Compensation impacts would generally remain indirect.

Scenario 3: Navigation or trajectory-model failure affecting Helionis and Oryn

A third correlated scenario is a common-cause failure in autonomous navigation, trajectory prediction, or debris-mapping systems affecting Helionis and Oryn at the same time. Helionis contains asteroid clusters with irregular gravitational resonances, erratic drift patterns, frequent micro-collisions, and shifting debris clouds. Oryn's asymmetric asteroid ring creates localised zones of rapid orbital shear and fluctuating gravitational gradients, and mining activity is expanding into these hazardous outer zones. A shared model error, software defect, or corrupted orbital update could therefore generate simultaneous losses in both systems.

This scenario is especially relevant for Cargo Loss, Equipment Failure, and Business Interrup-

tion. Workers' Compensation may be affected indirectly if manual intervention, emergency repairs, or evacuation activity is required.

Scenario 4: Inter-system transport congestion, debris cascade, or regulatory shock affecting all three systems

A fourth and broader correlated scenario affects all three systems through the company's transport network and regulatory exposure. Cosmic Quarry has invested heavily in port terminals, inter-system transportation links, and self-sustaining habitats, and has already faced environmental disputes involving debris generation and orbital saturation from transports between solar systems. A transport collision, fragmentation event, or debris cascade could therefore trigger route restrictions, inspection holds, orbital traffic throttling, and broader regulatory scrutiny across the entire interstellar network.

This scenario is expected to have the broadest loss footprint, spanning Cargo, Business Interruption, contingent Equipment Failure, and potentially liability-related costs. Workers' Compensation impacts may also arise indirectly if congestion, delay, or emergency operational changes create additional staff exposure.

Hazard-by-scenario summary matrix

Table E.1: Hazard-level interpretation of correlated scenarios

Scenario	Systems	WC	Cargo	EF	BI	Notes
Deep-space radiation transient	Bayesia + Oryn	L	H	H	H	Damages electronics, shielding, and remote extraction continuity.
Relay / communications backbone failure	Bayesia + Oryn	L	H	M	H	Disrupts routing, diagnostics, repairs, and operating continuity.
Navigation / trajectory-model failure	Helionis + Oryn	L-M	H	H	H	Shared model error amplified by hazardous orbital conditions.
Transport congestion / debris / regulatory shock	All three	M	H	M	H	Broad logistics and compliance disruption across systems.

Notes: WC = Workers' Compensation; EF = Equipment Failure; BI = Business Interruption.

L = Low, M = Medium, H = High.

F Macroeconomic forecasting inputs

To support pricing, capital modeling, and discounting, we project inflation and interest rates over the next 10 years using a parsimonious time-series model. The projected variables are the annual inflation rate, the overnight bank lending rate, the 1-year risk-free spot rate, and the 10-year risk-free spot rate.

Because the historical annual sample is short, we use a $VAR(1)$ specification to capture the joint dynamics among these variables while avoiding overfitting. To prevent economically implausible negative forecasts, each modeled variable is expressed as a logarithmic deviation from a positive long-run anchor. Specifically, for each variable x_t , we define

$$z_t = \log \left(\frac{x_t}{\bar{x}} \right),$$

where $\bar{x}$ is the long-run anchor.

The anchors are constructed as exponentially weighted averages of the most recent 15 years of observations, with larger weights assigned to more recent years. This allows the forecast to remain data-driven while reflecting recent macro-financial conditions more strongly than older observations.

The transformed variables are then modeled jointly as

$$Z_t = c + AZ_{t-1} + \varepsilon_t,$$

where Z_t collects the four log-gap variables. Forecasts are generated recursively for the next 10 years and then converted back to levels by

$$\hat{x}_{t+h} = \bar{x} \exp(\hat{z}_{t+h}).$$

This transformation ensures that forecasted inflation and nominal interest rates remain strictly positive. After forecasting the 1-year and 10-year spot rates, the term spread is calculated as

$$\widehat{\text{Spread}}_{t+h} = \hat{r}_{t+h}^{10Y} - \hat{r}_{t+h}^{1Y}.$$

Table F.1 reports the resulting baseline macroeconomic forecast used in pricing and capital modeling.

Table F.1: Baseline 10-year macroeconomic forecast

Year	Inflation	Overnight	1Y Rate	10Y Rate	Spread
2175	0.0090	0.0262	0.0253	0.0348	0.0095
2176	0.0065	0.0109	0.0116	0.0251	0.0136
2177	0.0076	0.0058	0.0069	0.0212	0.0144
2178	0.0107	0.0047	0.0058	0.0208	0.0150
2179	0.0148	0.0051	0.0064	0.0222	0.0158
2180	0.0178	0.0064	0.0078	0.0241	0.0164
2181	0.0186	0.0079	0.0093	0.0257	0.0164
2182	0.0178	0.0088	0.0102	0.0264	0.0162
2183	0.0165	0.0091	0.0104	0.0263	0.0159
2184	0.0155	0.0088	0.0100	0.0259	0.0158

Overall, the forecast suggests a temporary softening in inflation and short-term interest rates,

followed by gradual normalization. These projections are used as baseline macroeconomic inputs rather than precise point predictions.

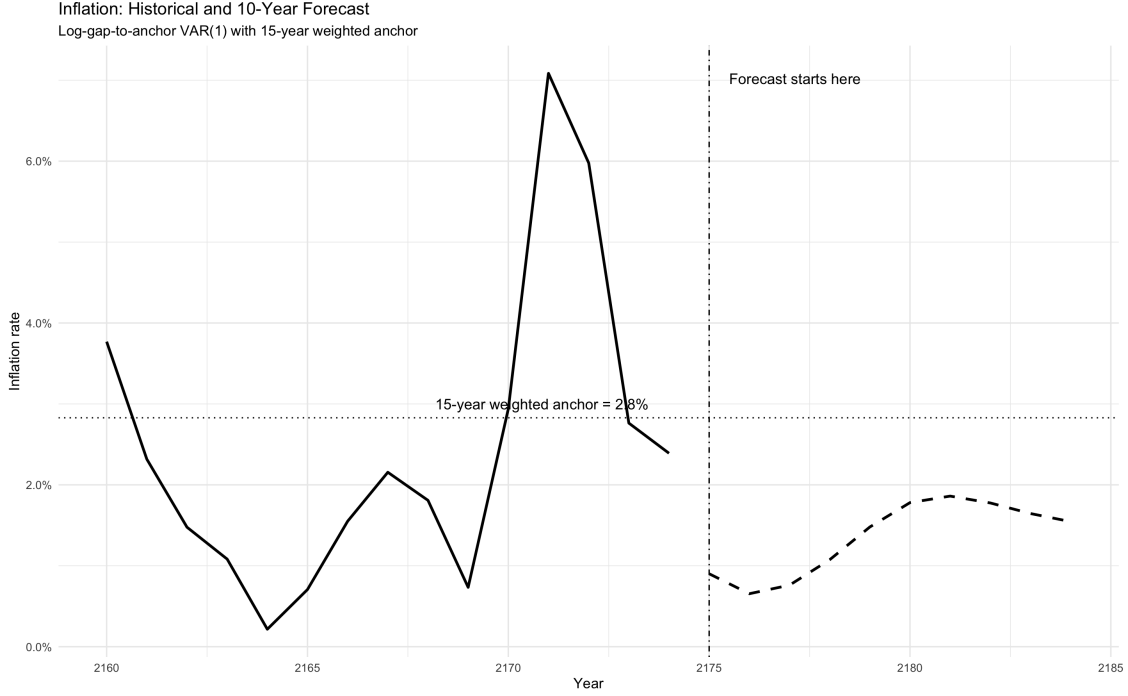


Figure F.1: Inflation: Historical and 10-Year Forecast

F.1 Yield curve construction at 2175

For pricing at the initial underwriting date of 2175, we require a term structure of risk-free spot rates for maturities from 1 to 10 years. The baseline macroeconomic forecast provides the 1-year and 10-year risk-free spot rates for 2175:

$$r_{2175,1} = 0.0253, \quad r_{2175,10} = 0.0348.$$

Because intermediate spot rates are not directly forecast, we construct the 2175 yield curve by linear interpolation. For maturity $m = 1, \dots, 10$, the interpolated spot rate is

$$r_{2175,m} = r_{2175,1} + \frac{m-1}{9} (r_{2175,10} - r_{2175,1}).$$

This implies a positively sloped yield curve in 2175, consistent with the positive term spread reported in Table F.1. The resulting maturity-specific spot rates are used to discount future expected losses, premiums, and net cash flows for long-term contracts issued at time 0.

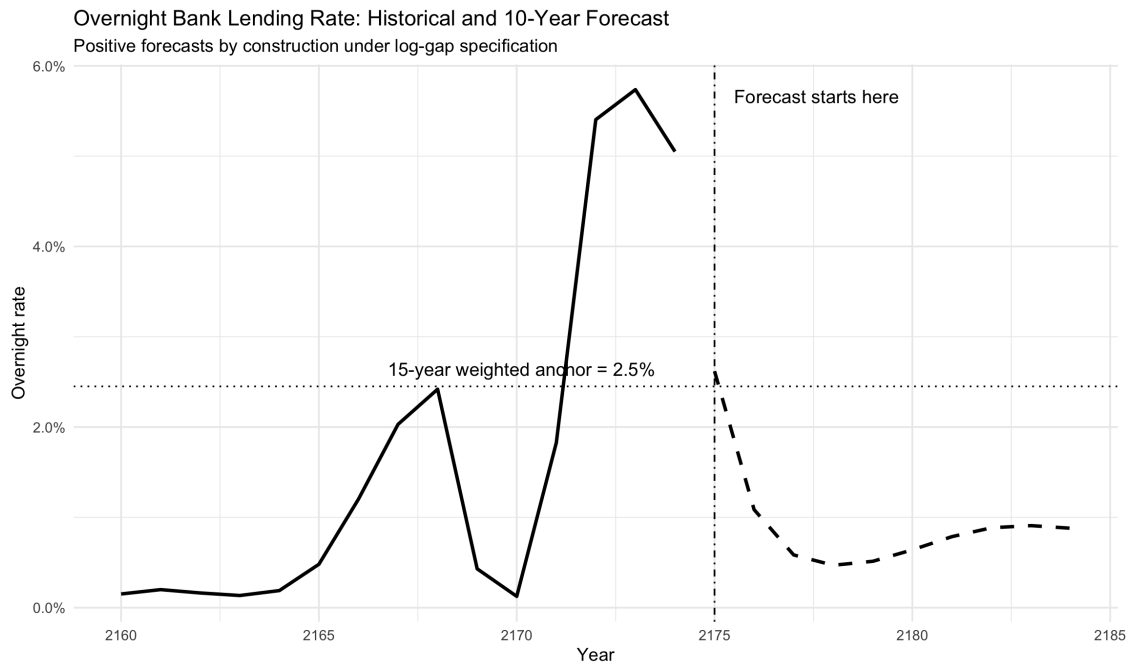


Figure F.2: Overnight Bank Lending Rate

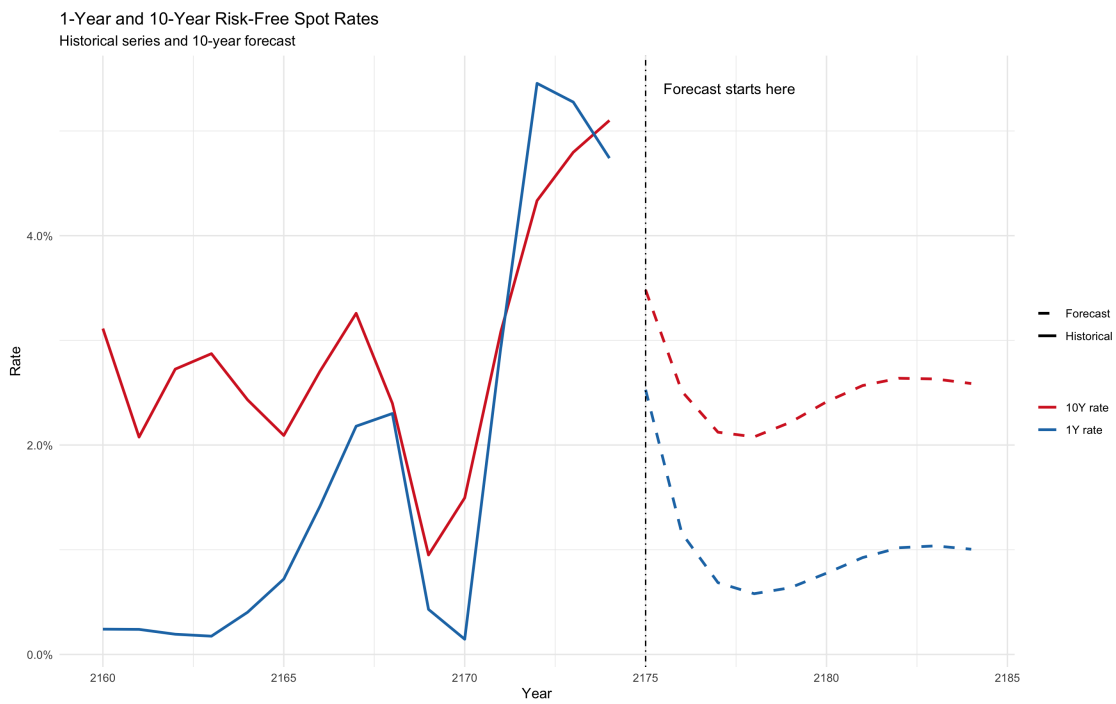


Figure F.3: 1-Year and 10-Year Risk-Free Spot Rates

Table F.2: Interpolated 2175 risk-free spot rates and discount factors

Maturity m	Spot rate $r_{2175,m}$	Discount factor $(1 + r_{2175,m})^{-m}$
1	0.0253	0.9753
2	0.0264	0.9493
3	0.0274	0.9221
4	0.0285	0.8938
5	0.0295	0.8646
6	0.0306	0.8347
7	0.0316	0.8041
8	0.0327	0.7731
9	0.0337	0.7418
10	0.0348	0.7103

For the one-year product issued in 2175, only the one-year spot rate $r_{2175,1}$ is required. For the ten-year product issued in 2175, the full set of maturity-specific spot rates $\{r_{2175,m}\}_{m=1}^{10}$ is used.

G Data cleaning summary

Before model fitting, all historical claims datasets were cleaned using a consistent rule-based procedure. Frequency and severity data were processed separately, but the same general principles were applied across all datasets. Table G.1 summarises the main cleaning steps.

Table G.1: Summary of data cleaning procedure

Cleaning step	Description
Data import	Raw Excel data were imported and separated into frequency and severity datasets for subsequent cleaning and modelling.
String standardisation	Inconsistent text labels were standardised, including removal of invalid suffixes and alignment of category names with the data dictionary.
Type conversion	Categorical variables were converted to factor type, while quantitative variables were converted to numeric format before validation checks.
Range validation	Numerical fields were checked against admissible data-dictionary ranges, and values outside the valid range were recoded as missing. The main exception was <code>claim.amount</code> for each hazard line: although the data dictionary provided reference intervals, we did not impose an upper-bound filter and only removed non-positive values, in order to preserve large-loss observations for severity and tail-risk modelling.
Missing-value review	Missing values were summarised by variable, and incomplete observations were removed using complete-case filtering prior to model fitting.
Duplicate handling	Duplicate records were identified using the relevant record identifier, and only the first occurrence was retained.
Category cleanup	Empty factor levels were dropped after filtering to ensure consistency in the final modelling dataset.
Quality checks	Summary counts and descriptive checks were performed after cleaning to confirm the plausibility and consistency of the remaining data.

We adopted a conservative cleaning principle: implausible observations were recoded to missing and excluded rather than imputed, in order to avoid introducing artificial structure into the claims data used for actuarial modelling.

H Detailed ESG scoring framework

This appendix presents the detailed methodology used to construct the ESG score for Cosmic Quarry. The score is designed as a transparent company-assessment tool rather than a direct pricing variable for insured losses. It is intended to summarize the firm’s controversy exposure, operating conditions, and governance quality in a way that is observable to the insurer and useful for underwriting review.

H.1 Overall structure

For each solar system s , the total ESG score is defined as

$$ESG_s = 0.50E_s + 0.15S_s + 0.35G_s.$$

The weights are chosen to reflect the information set available in this case study. Environmental and governance considerations receive greater weight because the disclosed disputes are environmental in nature and also reflect governance failures. Social considerations receive a lower weight because direct employee-level data for Cosmic Quarry are unavailable.

The company-level ESG score is then obtained as an exposure-weighted average:

$$ESG_{\text{company}} = \omega_H ESG_H + \omega_B ESG_B + \omega_O ESG_O,$$

where the weights are constructed from current mine share, production share, and expansion share:

$$\omega_s = 0.4(\text{mine share}_s) + 0.4(\text{production share}_s) + 0.2(\text{expansion share}_s).$$

H.2 Exposure weights

Using the encyclopedia data on mine counts, production, and planned expansion, the weights are:

Table H.1: Exposure weights used in the company-level ESG aggregation

Solar system	Mine share	Production share	Expansion share	ω_s
Helionis	30/55	375/750	25/65	0.495
Bayesia	15/55	250/750	25/65	0.319
Oryn	10/55	125/750	15/65	0.186

The 0.2 weight on expansion share is intentionally smaller than the weights on mine share and production share. Current footprint and current operating scale should dominate the exposure measure, while expansion should still enter as a forward-looking source of additional regulatory friction, habitat creation, logistics intensity, and environmental controversy risk.

H.3 Environmental pillar

The environmental pillar is defined as

$$E_s = 0.35E_{1,s} + 0.30E_{2,s} + 0.20E_{3,s} + 0.15E_{4,s},$$

where:

$E_{1,s}$: physical environmental hazard,

$E_{2,s}$: transport, debris, and controversy-channel exposure,

$E_{3,s}$: current footprint intensity,

$E_{4,s}$: mitigation and control infrastructure.

H.3.1 $E_{1,s}$: physical environmental hazard

This component captures the natural and operating environment of each solar system. It is scored directly from the descriptive text.

Table H.2: Mapping from descriptive statements to physical environmental hazard scores

Solar system	Key descriptive signals	Score
Helionis	Stable main-sequence star, low flare activity, predictable solar wind, mild seasonal variation, manageable atmosphere, minimal tectonic activity, geologically stable mineral veins. These signals indicate a relatively stable natural environment.	70
Bayesia	Temporary radiation spikes, high gravity, thin magnetosphere, elevated ambient radiation, and frequent temperature extremes. These indicate materially harsher physical conditions.	40
Oryn	Low visibility, potential sporadic flares, asymmetric asteroid ring, orbital shear, fluctuating gravitational gradients, rugged terrain, and tectonic scars. These indicate the harshest physical environment among the three systems.	25

H.3.2 $E_{2,s}$: transport, debris, and controversy-channel exposure

This component maps the descriptive statements most closely related to the disclosed dispute channels, namely forward contamination, landscape alteration, debris generation, and orbital

saturation.

Table H.3: Mapping from descriptive statements to transport/debris exposure scores

Solar system	Key descriptive signals	Score
Helionis	Erratic drift patterns, frequent micro-collisions, shifting debris clouds, and spatial clutter after fragmentation events imply meaningful exposure to debris-related controversy channels.	40
Bayesia	Heavily mapped asteroid belt, stable orbits, efficient navigation, standardized supply routes, orbital stations, and reliable relays indicate comparatively strong control of logistics and transport risk.	75
Oryn	Outward expansion into hazardous zones, asymmetric asteroid ring, rapid orbital shear, and fluctuating gravitational gradients indicate elevated transport and orbital controversy risk.	25

H.3.3 $E_{3,s}$: current footprint intensity

This component is calculated mechanically as

$$E_{3,s} = 100 (1 - FI_s) ,$$

where

$$FI_s = 0.4(\text{mine share}_s) + 0.4(\text{production share}_s) + 0.2(\text{expansion share}_s).$$

The resulting scores are:

Table H.4: Footprint-intensity calculation

Solar system	FI_s	$E_{3,s}$
Helionis	0.4951	50.5
Bayesia	0.3193	68.1
Oryn	0.1855	81.5

A lower footprint-intensity score indicates larger current scale and therefore greater environmental and regulatory pressure.

H.3.4 $E_{4,s}$: mitigation and control infrastructure

This component captures the extent to which the company has already built infrastructure that reduces environmental controversy risk.

Table H.5: Mapping from descriptive statements to mitigation/control scores

Solar system	Key descriptive signals	Score
Helionis	Company-wide habitats, port terminals, transportation networks, relay satellites, and long-standing mining operations support a reasonably strong mitigation environment.	70
Bayesia	Standardized routes, established orbital stations, reliable relays, and the same company-wide infrastructure imply the strongest mitigation capacity of the three systems.	80
Oryn	Amplified beacons, relay drones, and improving contact between hubs and deep-extraction vessels provide mitigation, but infrastructure remains rapidly evolving and communications remain inconsistent.	55

H.3.5 Environmental pillar results

$$E_H = 0.35(70) + 0.30(40) + 0.20(50.5) + 0.15(70) = 58.2$$

$$E_B = 0.35(40) + 0.30(75) + 0.20(68.1) + 0.15(80) = 61.4$$

$$E_O = 0.35(25) + 0.30(25) + 0.20(81.5) + 0.15(55) = 46.4$$

H.4 Social pillar

Because direct employee-level data are not available for Cosmic Quarry, the social pillar is interpreted as a combination of workforce-condition harshness and protection readiness rather than realized labor outcomes. We define

$$S = \min \{65, 0.50S_1 + 0.30S_2 + 0.20S_3\},$$

where:

- S_1 : workforce environment harshness,
- S_2 : response and support conditions,
- S_3 : people-protection readiness.

H.4.1 S_1 : workforce environment harshness

Table H.6: Mapping from descriptive statements to workforce-harshness scores

Solar system	Key descriptive signals	Score
Helionis	Manageable atmosphere, mild seasonal variation, minimal tectonic activity, and long-standing operations imply the mildest workforce environment of the three systems.	70
Bayesia	High gravity, elevated radiation, thin magnetosphere, and frequent temperature extremes imply a materially harsher working environment.	35
Oryn	Low visibility, rugged terrain, outward expansion into hazardous zones, and inconsistent long-range communications imply the harshest workforce environment.	25

H.4.2 S_2 : response and support conditions

Table H.7: Mapping from descriptive statements to support/readiness scores

Solar system	Key descriptive signals	Score
Helionis	Relay satellites, long-standing mining operations, and company-wide habitats provide a moderate-to-strong support environment.	70
Bayesia	Standardized supply routes, orbital stations, reliable relays, and broader infrastructure investment imply the strongest support and response setting.	80
Oryn	Beacons, relay drones, and growing hub-to-vessel contact improve support, but communications remain inconsistent and infrastructure is still evolving.	55

H.4.3 S_3 : people-protection readiness

This component is inferred from company-wide signals, especially the presence of approximately 4,300 staff monitoring environmental, safety, and health issues, together with large investments in habitats and transport infrastructure. Because no system-specific training, hours-worked, PPE, or supervision data are available, we assign a common readiness score of 65 across the three systems.

H.4.4 Social pillar results

$$S_H = \min\{65, 0.50(70) + 0.30(70) + 0.20(65)\} = 65.0$$

$$S_B = 0.50(35) + 0.30(80) + 0.20(65) = 54.5$$

$$S_O = 0.50(25) + 0.30(55) + 0.20(65) = 42.0$$

H.5 Governance pillar

The governance pillar is defined as

$$G_s = 0.25G_1 + 0.25G_2 + 0.25G_3 + 0.25G_4,$$

where:

G_1 : formal governance structure,

G_2 : oversight intensity,

G_3 : system execution and control maturity,

G_4 : controversy and penalty history.

H.5.1 G_1 : formal governance structure

We assign a common score of 80 to all systems based on the fact that Cosmic Quarry is a listed public company with a board of directors, executive and non-executive members, and a management structure consolidated under 12 executives.

H.5.2 G_2 : oversight intensity

This component is based on the monitoring-staff ratio:

$$\frac{4300}{35809} = 12.01\%.$$

The total number of employees is obtained by summing all employee counts reported in the personnel dataset.

We apply the following rule:

Monitoring ratio	Score
< 5%	40
5% – 10%	60
10% – 15%	75
15% – 20%	85
> 20%	95

Since 11.94% lies in the 10%–15% band, we set $G_2 = 75$.

H.5.3 $G_{3,s}$: system execution and control maturity

Table H.8: Mapping from descriptive statements to execution/control maturity scores

Solar system	Key descriptive signals	Score
Helionis	Long-standing operations and relay infrastructure indicate reasonably mature execution, although fragmentation clutter reduces control smoothness.	68
Bayesia	Standardized supply routes, mapped orbits, established orbital stations, and reliable relays indicate the strongest operating control environment.	80
Oryn	Some control systems exist, including beacons and relay drones, but infrastructure is still evolving and communications remain inconsistent.	55

H.5.4 G_4 : controversy and penalty history

We score controversy mechanically as

$$G_4 = 100 - 25(\text{fine event}) - 20(\text{settlement event}) - 10(\text{repeat-event flag}).$$

Since the encyclopedia discloses one fine event, one settlement event, and repeat controversy over time, we obtain

$$G_4 = 100 - 25 - 20 - 10 = 45.$$

H.5.5 Governance pillar results

$$G_H = 0.25(80) + 0.25(75) + 0.25(68) + 0.25(45) = 67.0$$

$$G_B = 0.25(80) + 0.25(75) + 0.25(80) + 0.25(45) = 70.0$$

$$G_O = 0.25(80) + 0.25(75) + 0.25(55) + 0.25(45) = 63.8$$

H.6 Final ESG scores

Combining the three pillars yields:

$$ESG_H = 0.50(58.2) + 0.15(65.0) + 0.35(67.0) = 62.3$$

$$ESG_B = 0.50(61.4) + 0.15(54.5) + 0.35(70.0) = 63.4$$

$$ESG_O = 0.50(46.4) + 0.15(42.0) + 0.35(63.8) = 51.8$$

The company-level ESG score is

$$ESG_{\text{company}} = 0.495(62.3) + 0.319(63.4) + 0.186(51.8) = 60.7.$$

Table H.9: Final ESG scorecard

Solar system	E_s	S_s	G_s	ESG_s
Helionis	58.2	65.0	67.0	62.3
Bayesia	61.4	54.5	70.0	63.4
Oryn	46.4	42.0	63.8	51.8
Company overall	–	–	–	60.7

H.7 Interpretation and underwriting use

The resulting ESG scores should be interpreted as underwriting-reference measures rather than direct rating variables for insured losses.

A practical interpretation is:

ESG range	Underwriting interpretation
70+	Generally acceptable; ESG used as background information
55–70	Insurable, but subject to closer review and monitoring
40–55	Cautious or conditional underwriting
Below 40	Unfavorable without substantial remediation

Under this interpretation, Helionis and Bayesia fall into the insurable-but-monitored range, whereas Oryn is materially closer to the cautious-underwriting range.

Finally, we emphasize that this ESG framework is not intended to replace actuarial pricing for the covered insurance lines. Rather, it is used to summarize the firm’s environmental, social, and governance profile, support underwriting judgment, and assess potential future controversy and penalty exposure.

I Cosmic Quarry Financial Performance Forecast

To assess the financial sustainability of Cosmic Quarry Mining Corporation over the long-term horizon, we project key financial indicators based on the company’s anticipated operational expansion and productivity trends described in the project data. The projection reflects a simplified financial outlook over a 10-year period and is used primarily to understand the potential scale and stability of the insured’s operations.

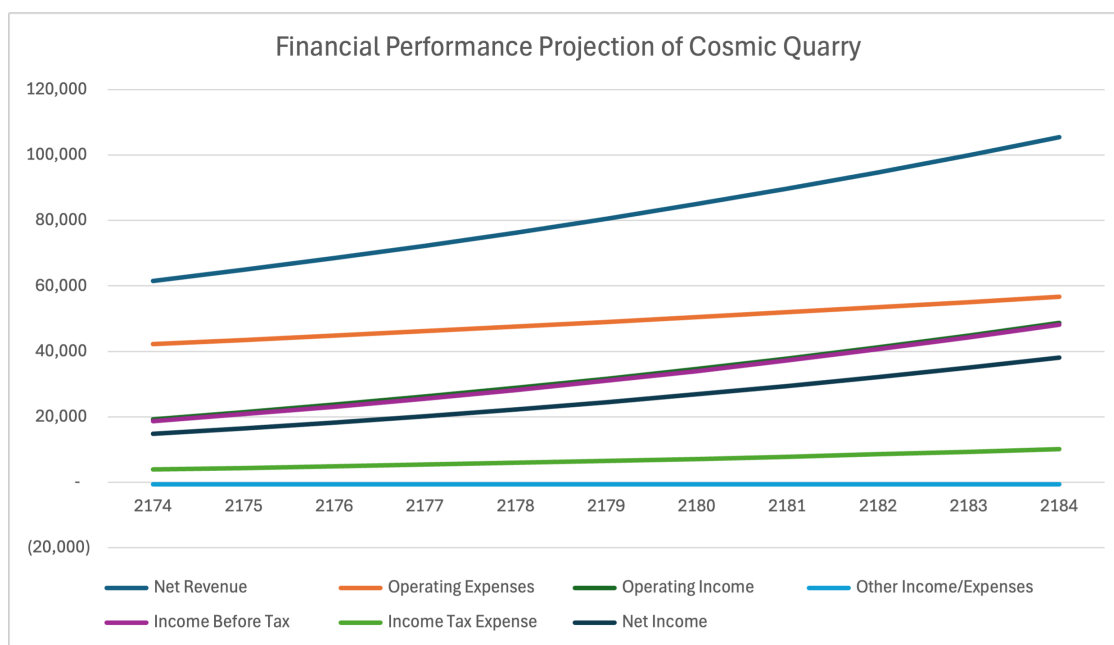


Figure I.1: Projected financial performance of Cosmic Quarry Mining Corporation

Figure I.1 illustrates the projected evolution of several financial metrics, including net revenue, operating expenses, operating income, income before tax, income tax expense, and net income. The projections indicate a steadily expanding mining operation, with revenue growth driven by increased production capacity and resource extraction across the three solar systems.

Operating expenses are assumed to grow at a slower rate than revenue due to operational scaling and efficiency improvements. As a result, operating income and net income are projected to increase over time, suggesting improving profitability as the mining operations mature.

This financial outlook provides context for the insurance portfolio design, as the scale of Cosmic Quarry's operations directly influences exposure levels for the four hazard coverages considered in this report. The projected growth trajectory therefore supports the long-term demand for insurance protection against operational disruptions, transportation losses, workforce incidents, and equipment failures.